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Text analytics in the legal domain:

the case of contracts and privacy policies in Claudette

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NEW COUNTOURS OF AI AND LAW

Recently, the popular perception of AI is that of **something** at the service of businesses

AI is currently affecting consumers'...

- privacy
- autonomy
- economic interests
- behaviour
- access to goods and services
- social exclusion

That does not have to be the case!
AI can unlock **consumer-empowering technologies**



How to empower consumers?

- **Protection** against unwanted monitoring (GDPR)
- **Support** in detecting unfair use of AI
- **Control** commercial practice fairness



“An opposing exercise of power is the principal solvent of economic power, the basic defense against its exercise in economic affairs”. Ken Galbraith

In the AI era an effective countervailing power needs to be supported by AI.

Meet CLAUDETTE.eui.eu



Automatically detect **potentially** unfair clauses in Terms of Services and Privacy Policies

- Consumers agree but don't read
- NGOs have competence to control but lack resources
- Business keeps using unlawful clauses

Terms of Service (ToS): The Training Set

The ToS Corpus

WHERE DID WE START?

... 50 ToS (manually annotated)...

7,090 sentences, 787 of which (11.1%) were labeled as positive, thus containing a potentially unfair clause.

WHERE ARE WE NOW?

... 100 ToS (manually annotated)...



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Part 1: Unfair Contract Terms Law and Practice

Directive 93/13 art 3.1:

A contractual term which has **not** been **individually negotiated** shall be regarded as **unfair** if, contrary to the requirement of good faith, it causes a **significant imbalance** in the parties' rights and obligations arising under the contract, to the detriment of the consumer.

Bottom-line: there are some types of clauses that traders are prohibited from using in the contracts.

9 unfairness categories (Art. 3 of Directive 93/13)

Type of clause	Symbol	# unfair clauses (142 Tos)	#documents (142 Tos)
Arbitration	<a>	156/165	54
Unilateral change	<ch>	506/506	138
Content removal	<cr>	261/261	102
Jurisdiction	<j>	180/218	118
Choice of law	<law>	192/225	132
Limitation of liability	<ltl>	971/1072	139
Unilateral termination	<ter>	624/624	134
Consent by using	<use>	370/370	129
Privacy included	<pinc>	115/115	80

Consent by using Clause

If a clause states that the consumer is bound by the terms of service simply by visiting the website or by downloading the app, or by using the service: **potentially unfair**

A **potentially unfair** consent by using clause (Airbnb):

```
<use2>By accessing or using the Airbnb Platform, you agree to comply with and be bound by these Terms of Service.</use2>
```

A **potentially unfair** consent by using clause (Facebook):

```
<use2>By using or accessing the Facebook Services, you agree to this Statement, as updated from time to time in accordance with Section 13 below.</use2>
```

Jurisdiction Clause

Where a dispute will be adjudicated?

If giving consumers a right to bring disputes in their place of residence: **clearly fair**

If stating that any judicial proceeding takes a residence away (i.e. in a different city, different country): **clearly unfair**

A clearly unfair jurisdiction clause (Dropbox):

<j3> You and Dropbox agree that any judicial proceeding to resolve claims relating to these Terms or the Services will be brought in the federal or state courts of San Francisco County, California, subject to the mandatory arbitration provisions below. Both you and Dropbox consent to venue and personal jurisdiction in such courts.</j3>

Limitation of Liability

For what actions/events the provider claims they will not be liable?

If stating that the provider may be liable: **clearly fair**

If stating that the provider will never be liable for any action taken by other people// damages incurred by the computer because of malware // When contains a blanket phrase like “to the fullest extent permissible by law”: **potentially unfair**

If stating that the provider will never be liable for physical injuries (health/life)// gross negligence// intentional damage: **clearly unfair**

Limitation of Liability

For what actions/events the provider claims they will not be liable?

A **clearly unfair** limitation of liability clause (Rovio):

<lttd3>In no event will Rovio, Rovio's affiliates, Rovio's licensors or channel partners be liable for special, incidental or consequential damages resulting from possession, access, use or malfunction of the Rovio services, [...] and, to the extent permitted by law, damages for **personal injuries**, [...] whether or not Rovio, Rovio's licensors or channel partners have been advised of the possibility of such damages.</lttd3>

Unilateral Termination

If the provider has the right to suspend and/ or terminate the service and/or the contract, and the **reasons** are specified => **potentially unfair**

If the provider has the right to suspend and/ or terminate the service and/or the contract, at any time for any or no reasons and/or without notice => **clearly unfair**

A **potentially unfair** <ter> clause (Dropbox):

<ter2>We reserve the right to suspend or terminate your access to the Services with notice to you if: (a) you're in breach of these Terms, (b) you're using the Services in a manner that would cause a real risk of harm or loss to us or other users, or (c) you don't have a Paid Account and haven't access.**</ter2>**

Unilateral Termination

If the provider has the right to suspend and/ or terminate the service and/or the contract, and the **reasons** are specified => **potentially unfair**

If the provider has the right to suspend and/ or terminate the service and/or the contract, at any time for any or no reasons and/or without notice => **clearly unfair**

A **clearly unfair** <ter> clause (Academia):

<ter3>Academia.edu reserves the right, at its sole discretion, to discontinue or terminate the Site and Services and to terminate these Terms, at any time and without prior notice.</ter3>

Privacy Included

Whenever a clause states (or it might be possible to assume) that the consumer consents to the privacy policy simply by using the service: **potentially unfair**

A potentially unfair clause (DeviantArt):

<use2>By using our Service, you agree to be bound by Section I of these Terms ("General Terms"), which contains provisions applicable to all users of our Service, including visitors to the DeviantArt website (the "Site").</use2> [...]

<pinc2>The terms of DeviantArt's privacy policy are incorporated into, and form a part of, these Terms.</pinc2>

An example from the Instagram Terms of Service

We reserve the right, in our sole discretion, to change these Terms of Use ("Updated Terms") from time to time.

Unless we make a change for legal or administrative reasons, we will provide reasonable advance notice before the Updated Terms become effective. You agree that we may notify you of the Updated Terms by posting them on the Service, and that your use of the Service after the effective date of the Updated Terms (or engaging in such other conduct as we may reasonably specify) constitutes your agreement to the Updated Terms.

<ltd2>TO THE EXTENT PERMITTED BY LAW, THE TOTAL LIABILITY OF GOOGLE, AND ITS SUPPLIERS AND DISTRIBUTORS, FOR ANY CLAIMS UNDER THESE TERMS, INCLUDING FOR ANY IMPLIED WARRANTIES, IS LIMITED TO THE AMOUNT YOU PAID US TO USE THE SERVICES (OR, IF WE CHOOSE, TO SUPPLYING YOU THE SERVICES AGAIN).</ltd2>

<ltd2>IN ALL CASES, GOOGLE, AND ITS SUPPLIERS AND DISTRIBUTORS, WILL NOT BE LIABLE FOR ANY LOSS OR DAMAGE THAT IS NOT REASONABLY FORESEEABLE.</ltd2>

Business uses of our Services

If you are using our Services on behalf of a business, that business accepts these terms. It will hold harmless and indemnify Google and its affiliates, officers, agents, and employees from any claim, suit or action arising from or related to the use of the Services or violation of these terms, including any liability or expense arising from claims, losses, damages, suits, judgments, litigation costs and attorneys' fees.

About these Terms

<ch2>We may modify these terms or any additional terms that apply to a Service to, for example, reflect changes to the law or changes to our Services. You should look at the terms regularly.</ch2> We'll post notice of modifications to these terms on this page. We'll post notice of modified additional terms in the applicable Service. Changes will not apply retroactively and will become effective no sooner than fourteen days after they are posted. However, changes addressing new functions for a Service or changes made for legal reasons will be effective immediately. If you do not agree to the modified terms for a Service, you should discontinue your use of that Service.

If there is a conflict between these terms and the additional terms, the additional terms will control for that conflict.

These terms control the relationship between Google and you. They do not create any third party beneficiary rights.

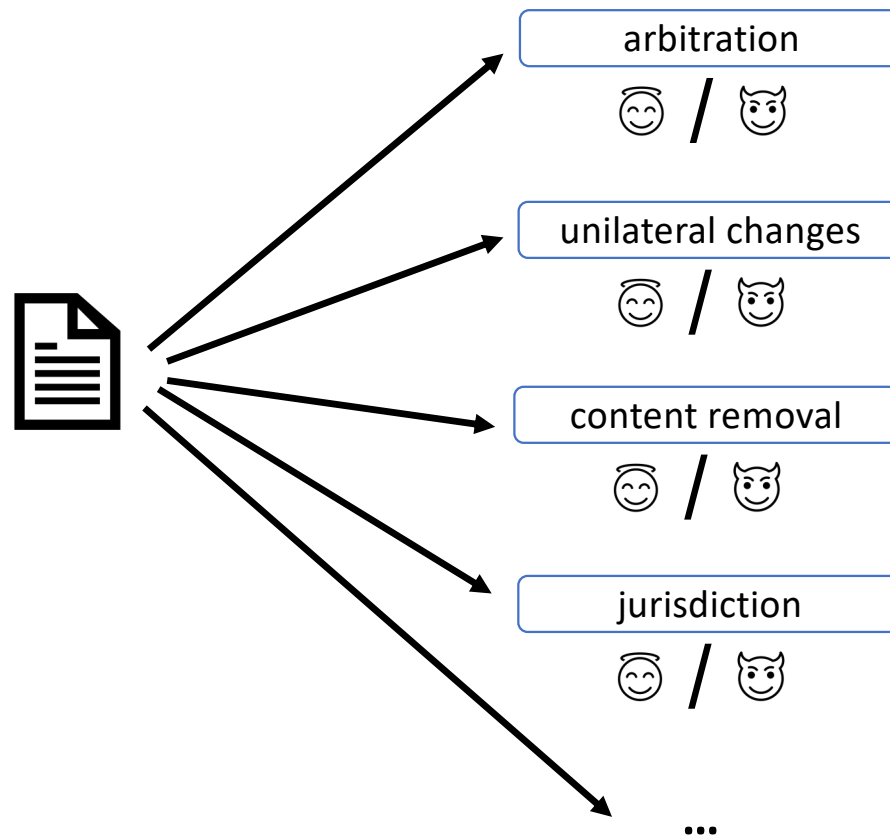
If you do not comply with these terms, and we don't take action right away, this doesn't mean that we are giving up any rights that we may have (such as taking action in the future).

If it turns out that a particular term is not enforceable, this will not affect any other terms.

<law2>The laws of California, U.S.A., excluding California's conflict of laws rules, will apply to any disputes arising out of or relating to these terms or the Services.</law2> <j3>All claims arising out of or relating to these terms or the Services will be litigated exclusively in the federal or state courts of Santa Clara County, California, USA, and you and Google consent to personal jurisdiction in those courts.</j3>

Task(s)

- For each clause,
- for each category,
- assess if it is fair or not



The beginning: Machine Learning Methodology

From a ML point of view, we modelled the problem as:

a **detection task**: does a sentence contain a potentially unfair clause? Positive (if p unfair), Negative (otherwise)

a **sentence classification task**: to what category does the unfair clause belong?

Approaches

- Bag of Words (BoW): leverages on the lexical information in sentences
- Tree kernels: leverages on grammatical structure of sentences
- Convolutional Neural Networks, SVM, etc.

Experiments

Leave-One-Out procedure: each document in turn, is used as test set, leaving the remaining documents for training set (4/5) and validation set (1/5) for model selection

3 Metrics

Precision: fraction of positive predictions, actually labelled as positive

$$= \frac{TP}{TP+FP}$$

Recall: fraction of positive examples that are correctly detected

$$= \frac{TP}{TP+FN}$$

F1: harmonic mean between precision and recall

$$= \frac{2PR}{P+R}$$

Baselines for comparison: random classifier

Experimental Results

Performance: Training set size = 50 Tos

Method	P	R	F_1
SVM—single model	0.729	0.830	0.769
SVM—combined model	0.798	0.782	0.781
Tree kernels	0.777	0.718	0.739
Convolutional neural networks	0.729	0.739	0.722
Long short-term memory networks	0.696	0.723	0.698
SVM-HMM—single model	0.759	0.778	0.758
SVM-HMM—combined model	0.859	0.687	0.757
Ensemble (C1+C2+C3+C6+C7)	0.826	0.797	0.805
Random baseline	0.125	0.125	0.125
Always positive baseline	0.123	1.000	0.217

Best performing: **Ensemble**

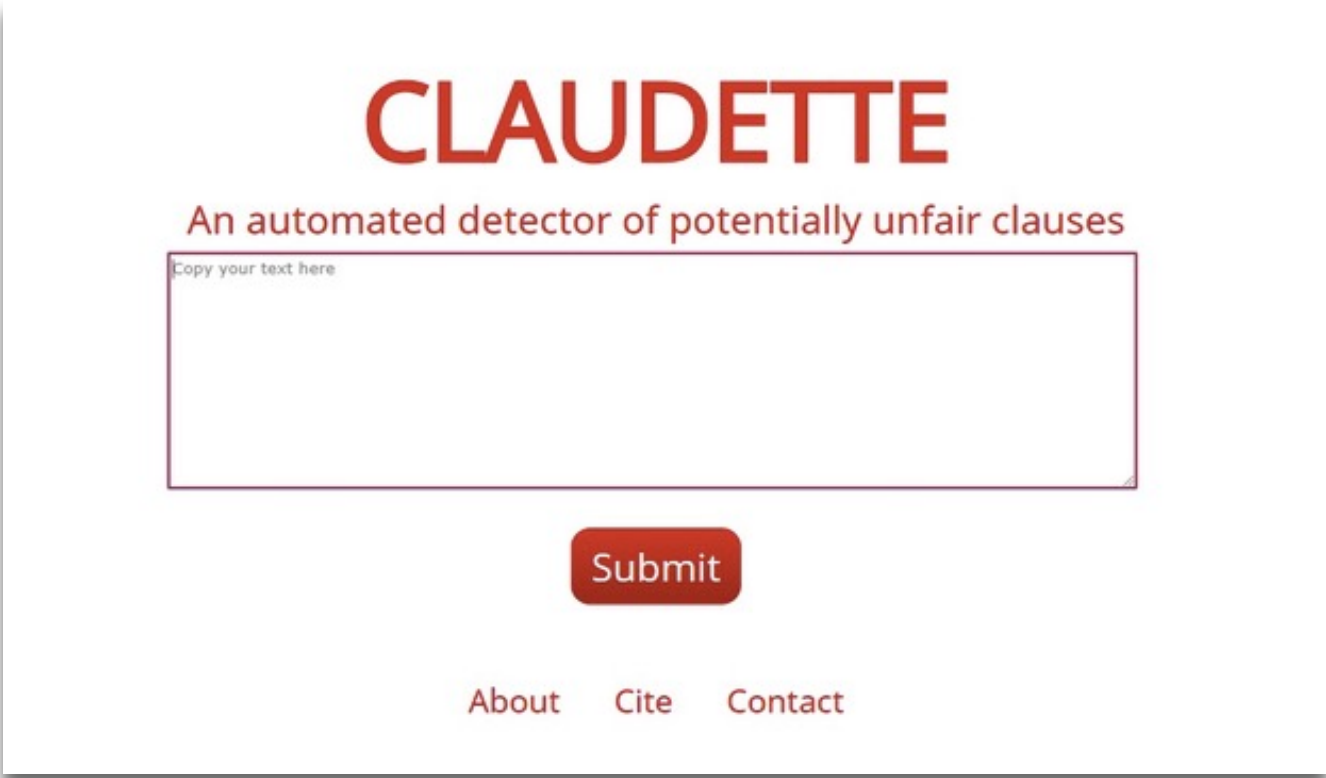
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Experimental Results

Claudette correctly detected around **80%** of the **potentially unfair** clauses in each category, ranging from **a minimum 72.7%** in the case of arbitration clauses, **up to 89.7%**, as in the case of jurisdiction clauses.

Tag	Precision	Recall	F ₁
Arbitration	0.832	0.814	0.823
Unilateral change	0.832	0.814	0.823
Content removal	0.713	0.780	0.745
Jurisdiction	1.000	0.941	0.970
Choice of law	0.984	0.886	0.932
Limitation of liability	0.961	0.905	0.932
Unilateral termination	0.786	0.932	0.853
Contract by using	0.949	0.957	0.953

An online server



The screenshot shows the CLAUDETTE web interface. At the top, the word "CLAUDETTE" is displayed in large, bold, red capital letters. Below it, the subtitle "An automated detector of potentially unfair clauses" is written in a smaller, dark red font. A large, empty rectangular text box with a thin red border is positioned below the subtitle, containing the placeholder text "Copy your text here" in the top-left corner. Centered below the text box is a red, rounded rectangular button with the word "Submit" in white. At the bottom of the interface, three links are listed in a dark red font: "About", "Cite", and "Contact".

An online server

CLAUDETTE

An automated detector of potentially unfair clauses

Copy your text here

[Submit](#)

[About](#) [Cite](#) [Contact](#)

CLAUDETTE

An Automated Detector of Potentially Unfair Clauses

Claudette found 4 potentially unfair clauses (displayed in **bold**) out of 5 sentences.
By hovering your cursor over each unfair sentence, you can see the most likely unfairness category.

[...]

Unilateral Change unfair clause

As such, the Services may change from time to time, at our discretion.

We may stop (permanently or temporarily) providing the Services or any features within the Services to you or to users generally.

We also retain the right to create limits on use and storage at our sole discretion at any time.

We may also remove or refuse to distribute any Content on the Services, suspend or terminate users, and reclaim usernames without liability to you.

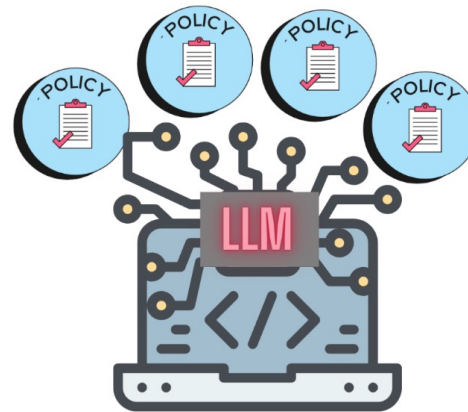
[Share link](#) [Save results](#)

[Try Again](#) [Contact](#)

New Method(s)

- Classical ML methods (e.g., SVM, NN)
- Transformer-based (e.g., Bert, Legal Bert) State of the art performance

RQ WHAT IF



Task: Detecting unfair clauses in ToS

Methods and Strategies



- **Single Prompt**
- **Multi-Prompt**
- **Pipeline**

Task: Detecting unfair clauses in ToS

Selection of examples in few-shot setting

➤ 8 examples for each category of unfair clauses

➤ Extracted from the training set and **hand-selected**

Criteria:

(i) coverage of **different market sectors**;

(ii) coverage of **diverse types** of **unfair** contracting **practices**, within each unfair-clause category;

(iii) length of clauses.

➤ 2 approaches: **long** vs **short** sentences

Selection of examples in few-shot setting

Approach I: long sentences

- **Ratio:** they are usually relevant for a diverse set of unfair practices.
- **Example:** a clause may limit or exclude the provider's liability by:
 - (i) **liability theory** (e.g., tort or contract law, strict/statutory/product liability, product liability); (ii) **causal link** with the damage (e.g., special, incidental, direct damages); (iii) **kind of damage** (e.g., economical, reputational); (iv) **standard of care** (e.g., negligence, gross negligence); (v) **cause of damage** (e.g., security breach, computer harms); and (vi) **compensation amount** (e.g., max 10 euro)
- **Advantages:**
 - wide range and of unfair practices
 - richer and more detailed context/scenarios, helping the generalization capabilities
- **Disadvantages:**,
 - SLMs input length constraints or difficulty in capturing long-range dependencies
 - SLMs struggle with processing lengthy sentences

Selection of examples in few-shot setting

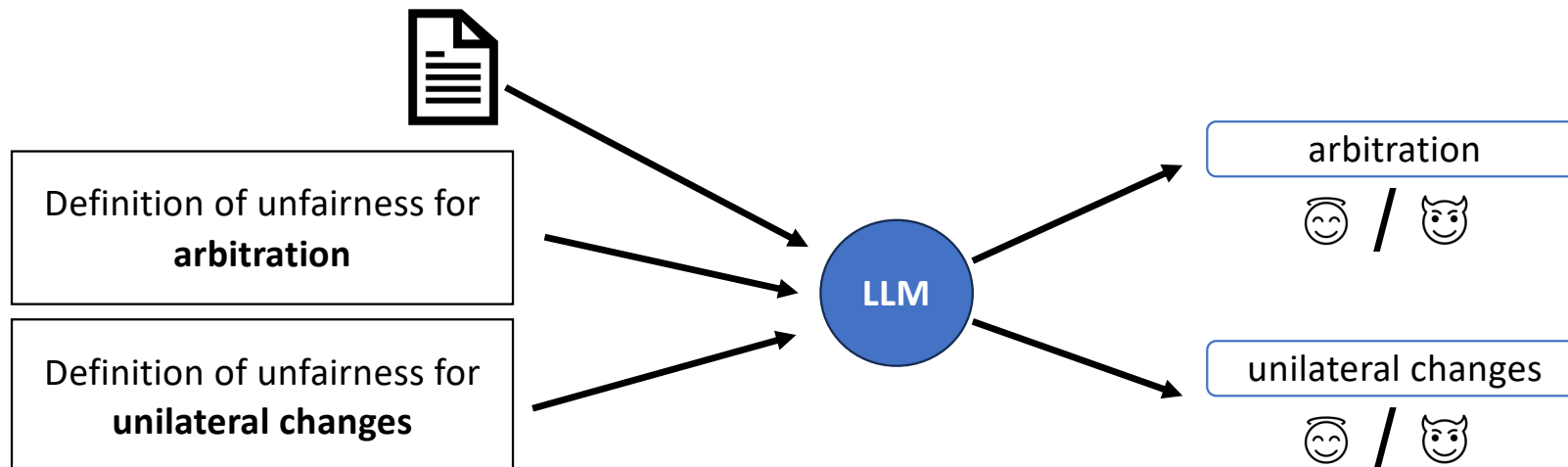
Approach II: short sentences

- Each clause is pertinent to one, or at most two, instances of unfair practice(s)
- **.Advantages:**
 - simpler to process and reduce the risk of truncation
 - allow models to focus on specific patterns
- **Disadvantages:**
 - Risk of oversimplification.
 - Reduce cohesiveness and create a misalignment with real world contractual language
 - Lose the broader context necessary to
 - Fully understand the relationships between different elements.

Generative AI Approaches

Single Prompt

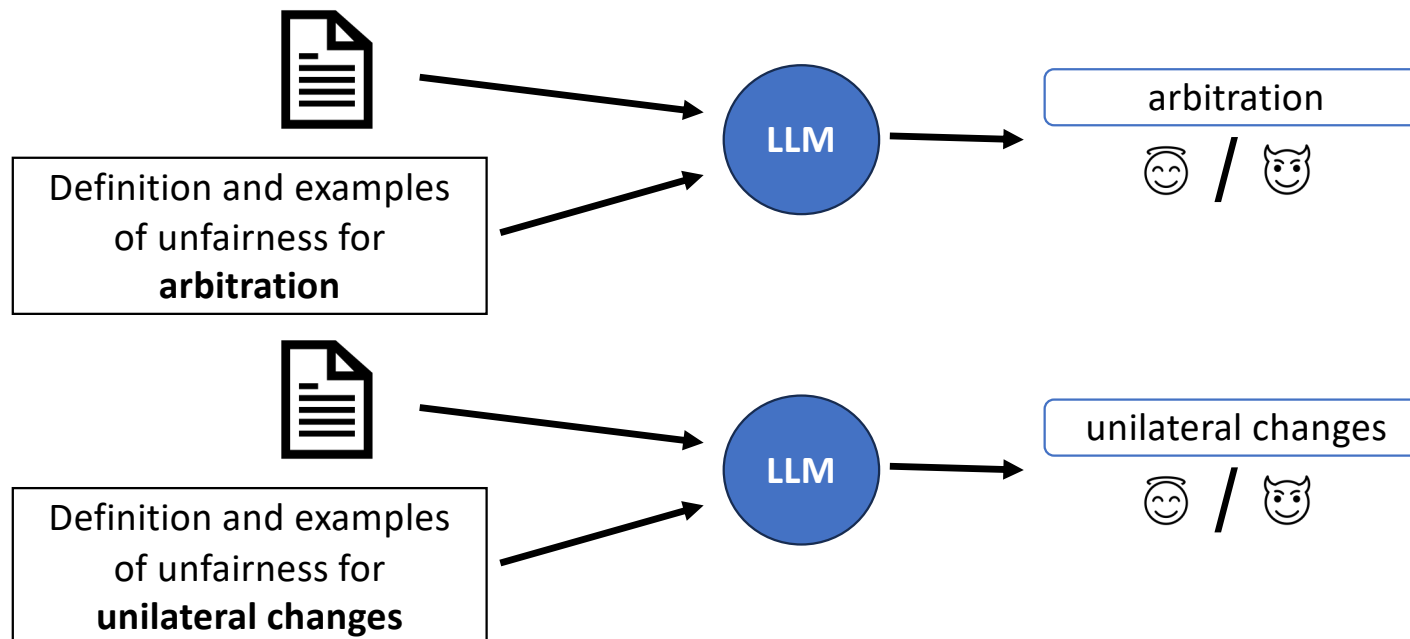
- **Detection and Classification performed at once**
- **Comprehensive prompt** with all the information
- **Not scalable** few-shot setting: all examples of all the categories in a single prompt, thus very large context!



Generative AI Approaches

Multi prompt

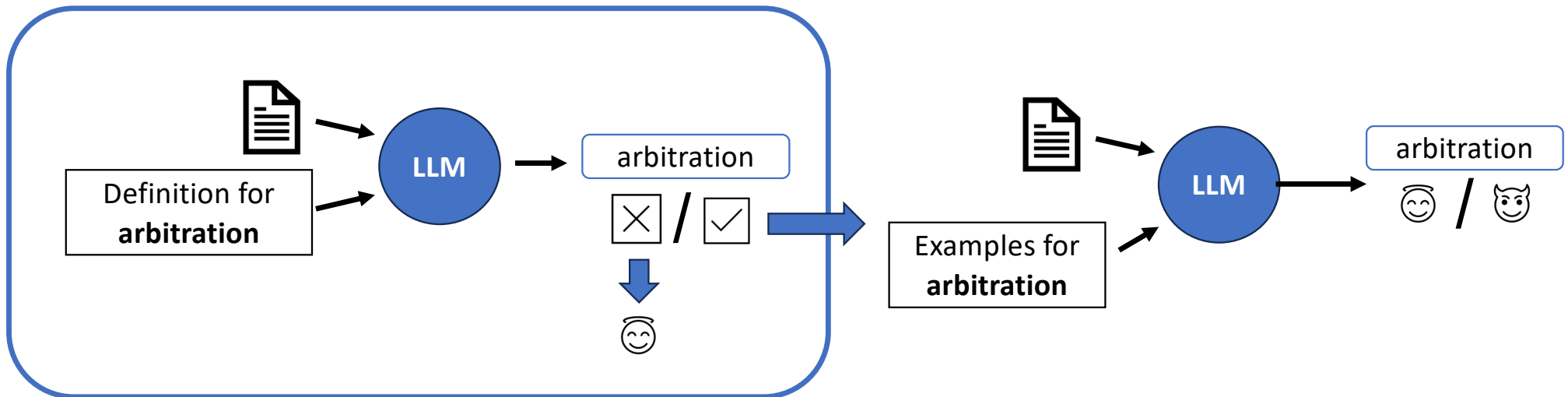
- One prompt per category
- Decomposition mitigates the risk of misunderstanding



Generative AI Approaches

Pipeline (1/2)

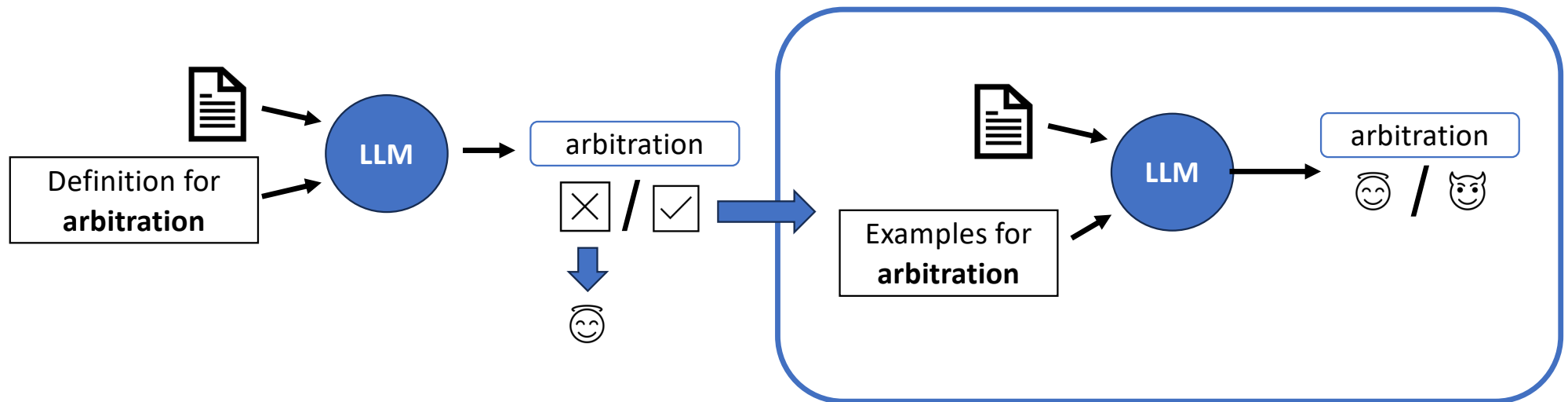
- Split the task into two subsequent steps
- First, assess whether the clause is about a certain category



Generative AI Approaches

Pipeline (2/2)

- Split the task into two subsequent steps
- Second, classify the clause as fair or unfair (only if it belongs to topic!)



Tested LLMs

Model	#Params	Context Length	Quantization
Qwen [22]	32B	128k	4 bit
Codestral [50]	22B	32k	4 bit
Phi3 [52]	14B	128k	8 bit
Nemo [51]	12B	128k	8 bit
Llama [13]	8B	128k	8 bit
Mistral [24]	7B	32k	8 bit
Law-Chat [8]	7B	128k	8 bit

Extended results

Model	Technique	Examples	Metrics			
			Micro-F1	Macro-F1	Macro Acc.	Macro Rec.
Llama3-8B	Single-prompt	/	0.71	0.37	0.38	0.61
Mistral-7B	Single-prompt	/	0.55	0.23	0.25	0.42
Law-Chat	Single-prompt	/	0.22	0.09	0.17	0.07
Nemo-12B	Single-prompt	/	0.72	0.39	0.42	0.57
Phi3-14B	Single-prompt	/	0.60	0.33	0.30	0.58
Codestral-22B	Single-prompt	/	0.15	0.19	0.37	0.17
Qwen-32B	Single-prompt	/	0.87	0.49	0.43	0.64
LLama3-8B	Multi-prompt - Zero shot	/	0.88	0.47	0.43	0.57
Mistral-7B	Multi-prompt - Zero shot	/	0.41	0.16	0.14	0.80
Law-Chat	Multi-prompt - Zero shot	/	0.83	0.39	0.33	0.62
Nemo-12B	Multi-prompt - Zero shot	/	0.61	0.26	0.20	0.81
Phi3-14B	Multi-prompt - Zero shot	/	0.79	0.41	0.31	0.77
Codestral-22B	Multi-prompt - Zero shot	/	0.34	0.21	0.19	0.86
Qwen-32B	Multi-prompt - Zero shot	/	0.88	0.61	0.56	0.75

- Single-prompt approach is not effective
- With multi-prompt we see several improvements
- Qwen clearly outperform alternatives

Extended results

Model	Technique	Examples	Metrics			
			Micro-F1	Macro-F1	Macro Acc.	Macro Rec.
Llama3-8B	Multi-prompt - Few shot	Long	0.79	0.40	0.30	0.84
		Short	0.78	0.40	0.31	0.78
Mistral-7B	Multi-prompt - Few shot	Long	0.53	0.24	0.20	0.85
		Short	0.52	0.22	0.18	0.87
Law-Chat	Multi-prompt - Few shot	Long	0.51	0.31	0.26	0.71
		Short	0.71	0.43	0.41	0.62
Nemo-12B	Multi-prompt - Few shot	Long	0.32	0.15	0.15	0.86
		Short	0.31	0.14	0.14	0.87
Phi3-14B	Multi-prompt - Few shot	Long	0.72	0.43	0.34	0.85
		Short	0.80	0.52	0.40	0.80
Codestral-22B	Multi-prompt - Few shot	Long	0.41	0.24	0.21	0.87
		Short	0.44	0.26	0.21	0.89
Qwen-32B	Multi-prompt - Few shot	Long	0.90	0.59	0.47	0.84
		Short	0.88	0.61	0.51	0.85

➤ Long prompts are rarely better than short prompts

Extended results

Model	Technique	Examples	Metrics			
			Micro-F1	Macro-F1	Macro Acc.	Macro Rec.
Llama3-8B	Pipeline	Long	0.90	0.58	0.51	0.71
		Short	0.91	0.55	0.54	0.61
Mistral-7B	Pipeline	Long	0.74	0.29	0.24	0.70
		Short	0.59	0.24	0.19	0.87
Law-Chat	Pipeline	Long	0.43	0.20	0.17	0.87
		Short	0.46	0.22	0.18	0.85
Nemo-12B	Pipeline	Long	0.92	0.62	0.57	0.72
		Short	0.94	0.62	0.71	0.59
Phi3-14B	Pipeline	Long	0.87	0.56	0.44	0.84
		Short	0.87	0.57	0.46	0.82
Codestral-22B	Pipeline	Long	0.80	0.45	0.35	0.85
		Short	0.80	0.47	0.36	0.85
Qwen-32B	Pipeline	Long	0.95	0.72	0.77	0.71
		Short	0.94	0.71	0.77	0.71

- Pipeline method improves almost all the performances
- Qwen in pipeline is the best performing model, but below BERT baselines

Comparison

Model	Technique	Category F1									
		fair	a	ch	cr	j	law	ltd	ter	use	pinc
Baselines											
Custom-Legalbert	Fine tuned	0.97	0.56	0.78	0.67	0.96	0.93	0.70	0.75	0.79	0.67
Deberta-base	Fine tuned	0.98	0.67	0.76	0.70	0.90	0.93	0.71	0.73	0.82	0.57
Small LLMs											
Mistral-7B	Pipeline-Long	0.85	0.07	0.19	0.08	0.12	0.1	0.3	0.38	0.14	0.20
LawChat	Multi Few-Short	0.87	0.36	0.32	0.08	0.65	0.24	0.19	0.55	<u>0.62</u>	0.38
Phi3-14B	Pipeline-Short	0.93	0.33	0.52	0.45	<u>0.74</u>	0.71	0.46	<u>0.61</u>	0.55	0.44
Llama3-8B	Pipeline-Long	0.95	<u>0.49</u>	<u>0.64</u>	0.55	0.68	0.54	<u>0.55</u>	0.58	0.42	0.36
Nemo-12B	Pipeline-Short	<u>0.97</u>	0.34	0.63	<u>0.64</u>	0.73	<u>0.80</u>	0.54	0.55	0.56	<u>0.45</u>
Large LLMs											
Qwen-32B	Pipeline-Long	0.98	0.48	0.70	0.64	0.91	0.93	0.67	0.57	0.65	0.69
Codestral-22B	Pipeline-Short	0.90	0.19	0.54	0.37	0.53	0.47	0.54	0.47	0.18	0.47

Computational Cost

- BERT models:
 - between 15-30 minutes for training
 - 1-2 ms for complete inference
- LLMs: depends on the setting.
 - Between 1 and 5 seconds, per inference.
- Transformer models are 1000 times faster than LLMs!
- The more an approach is “decomposed”, the more inference steps it requires, the more time is required.
 - Worst case for pipeline: 16 inferences for an instance!
- The size of the tested LLM does not seem to impact inference time strongly.

The Explainability development

CLAUDETTE

An Automated Detector of Potentially Unfair Clauses

Claudette found 3 potentially unfair clauses (displayed in **bold**) out of 295 sentences.
Below you can find a summary of the detected clauses, possibly linked to the most plausible rationales.

Potentially unfair clause #1

EXCEPT FOR CERTAIN TYPES OF DISPUTES MENTIONED IN THE ARBITRATION CLAUSE , YOU AND HEADSPACE AGREE THAT DISPUTES RELATING TO THESE TERMS OR YOUR USE OF THE PRODUCTS WILL BERESOLVED BY MANDATORY BINDING ARBITRATION , AND YOU WAIVE ANY RIGHT TO PARTICIPATE IN A CLASS-ACTION LAWSUIT OR CLASS-WIDE ARBITRATION .

Unfairness categories: **Arbitration**

[Hide/show rationales](#)

Potentially unfair clause #2

1.4 CHANGES TO TERMS Headspace reserves the right to change or update these Terms , or any other of our policies or practices , at any time , and will notify users by posting such changed or updated Terms on this page .

Unfairness categories: **Unilateral Change**

[Hide/show rationales](#)

Potentially unfair clause #3

Your continued use of the Products constitutes your agreement to abide by the Terms as changed .

Unfairness categories: **Contract by Using**

[Hide/show rationales](#)

CLAUDETTE

An Automated Detector of Potentially Unfair Clauses

Potentially unfair clause #1

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Unfairness categories: **Arbitration**

[Hide/show rationales](#)

Potentially unfair clause #2

1.4 CHANGES TO TERMS Headspace reserves the right to change or update these Terms , or any other of our policies or practices , at any time , and will notify users by posting such changed or updated Terms on this page .

Unfairness categories: **Unilateral Change**

[Hide/show rationales](#)

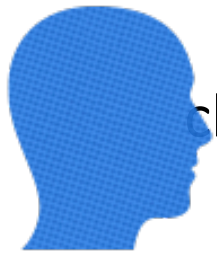
The clause is potentially unfair for **Unilateral Change** since the provider has the right for unilateral change of the contract, services, goods, features for any reason at its full discretion, at any time (score = 0.834)

Potentially unfair clause #3

Your continued use of the Products constitutes your agreement to abide by the Terms as changed .

Unfairness categories: **Contract by Using**

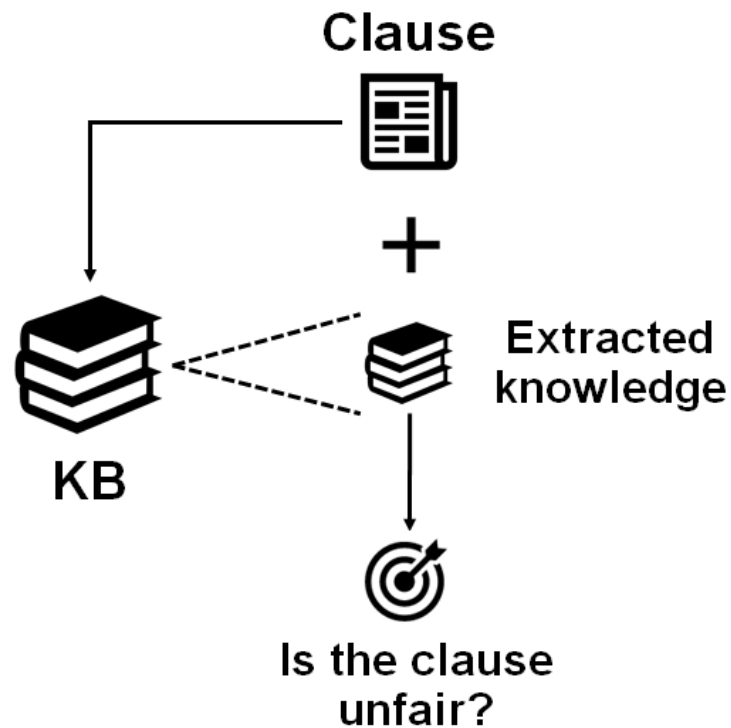
[Hide/show rationales](#)



Human Legal experts are able to recognize potentially unfair clauses thanks to their **background knowledge** of the domain.

- Rely on intuitions, trained on experience with relevant examples
- Able to explain their intuitions of unfairness, provide reasons why a clause is unfair (**Legal Rationales**), and use rationales to guide such intuitions
- Appealing to their background knowledge (e.g. Standards, Rules and Principles, Judicial precedents) as support for reasoning

How to improve CLAUDETTE with background knowledge?



Our new goal

Detect unfair clauses via knowledge checking

- Process this knowledge and **store** the information in some **memory**
- Understand **pieces of knowledge** relevant to a given **query**
- Retrieve **concepts** from memory
- Combine **memory and query** to make a prediction
- We rely on **MANNs** (RNN)

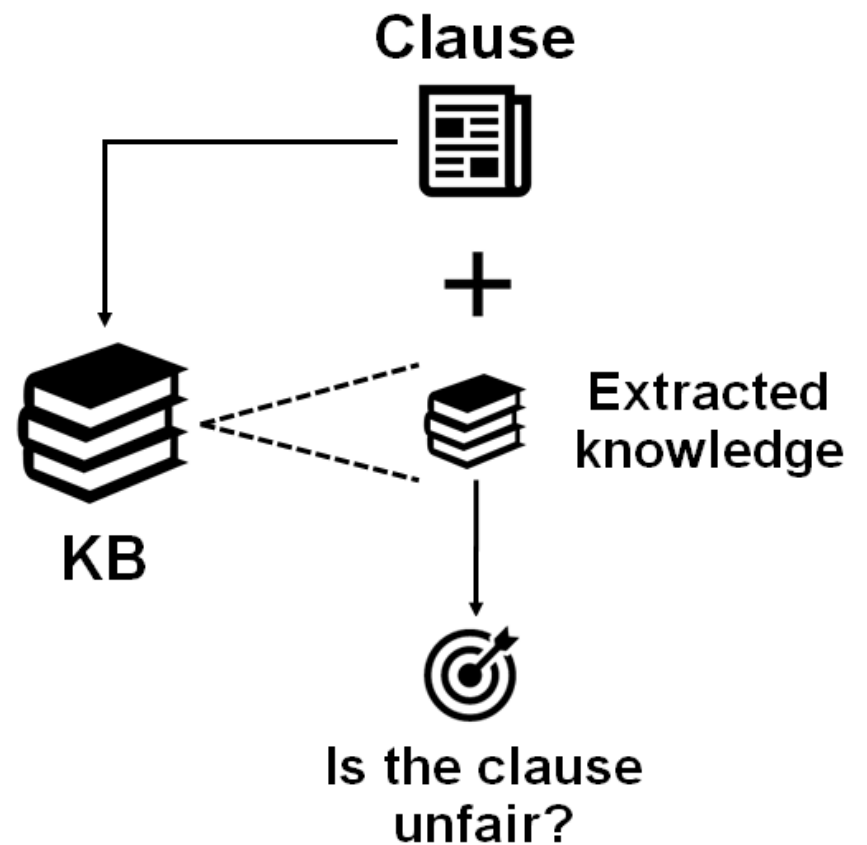
Exploiting Knowledge for Unfairness Identification

Our goal

Detect unfair clauses via knowledge checking

Requirements

- 👍 Input definition
- 👍 Knowledge definition
- 👍 Extraction mechanism



Knowledge base definition

- Focus on 100 ToS and 5 categories: (i) <ld>; (ii) <ch>; (iii) <ter>; (iv) <cr>; and (v) <a>
- The remaining 4 categories have been excluded since with the same types of practices and rationales.
- Novel structured corpus of different legal rationales
- Each rationale is identified through an ID
- Each clause can be matched (and thus indexed) with several potentially unfair practices/legal rationales (one-to-many mapping of clauses to legal rationales)
- The link between rationales and clauses can be used to instruct the system so that it can provide an explanation for the unfairness of particular clauses.



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Example: Liability exclusions and limitations

- **6 classes of clauses** limiting liability for damage on the basis of (i) liability theories, (ii) causal link, (iii) kind of damage, (iv) standard of care, (v) cause, (vi) compensation amount.
- Within each class, we distinguished a variety of potentially unfair practices, mapping **different questionable circumstances** under which ToS reduce or exclude the provider's liability
- **21 legal rationales** for (potentially) unfair ltd



Fig. 1 Conceptual map of LTD classes of (potentially) unfair clauses

Rationale examples: limitation of liability

		ID	Legal rationales
By kind	➤ Loss	dataloss	The provider is not liable for any disclosure, damage, destruction, corruption, failure to store or loss of data and material
		reputation	The provider is not liable for reputational and goodwill damages, loss
		anyloss	The provider is not liable for any loss , or broad categories of loss, resulting from the use of the service and or of the website, including lost profits, lost opportunity, lost business or lost sales, data loss, loss of goodwill, loss of reputation
		ecoloss	The provider is not liable for any loss of profits, loss of income, lost opportunity, lost business or lost sales, loss of revenue
	➤ Injury	injury	The provider is not liable for intentional offence and damage, physical or personal injury, including, emotional distress and death

Liability exclusions and limitations: examples

Example from the Box ToS (updated August 2021):

To the extent not prohibited by law, in no event will Box, its affiliates re-sellers, officers, employees, agents, suppliers or licensors be liable for: any direct, incidental, special, punitive, cover or consequential damages (including, without limitation, damages for lost profits, revenue, goodwill, use or content) however caused, under any theory of liability, including, without limitation, contract, tort, warranty, negligence or otherwise, even if Box has been advised as to the possibility of such damages.

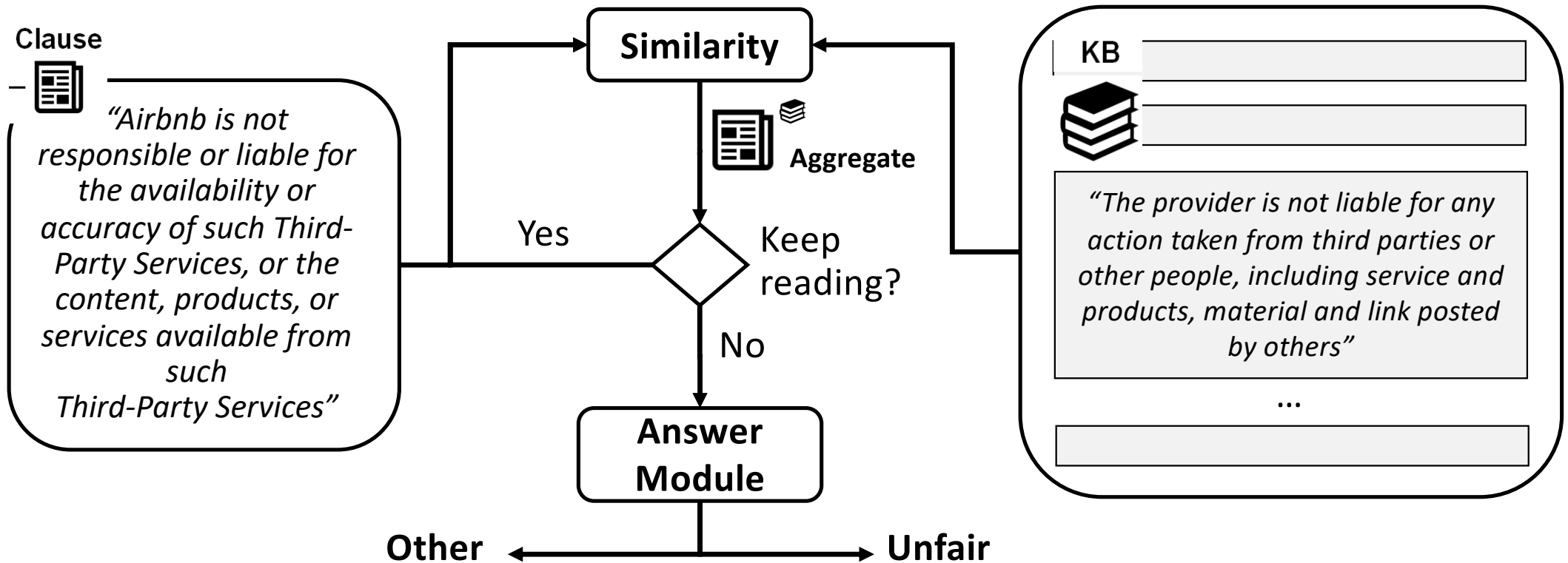
(IDs: **extent, anydamage, reputation, dataloss, ecoloss, awareness, theories**) .



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Exploiting Knowledge for Unfairness Identification

Esperimental Setup



Multilingualism: the German, Italian and Polish Claudette for ToS and PPs



Ein automatisierter Detektor für potenziell unlautere Klauseln
(GERMAN BETA VERSION)

Kopieren Sie Ihren Text hier

Submit

Creating a multilingual corpus

- A **multilingual parallel corpus**: 50 Terms of Service annotated in English, Italian, German and Polish (200 ToS in total).
- ToS retrieved from the Claudette **pre-existing corpus** (142 Tos)
- **Selection criteria**:
 - ❑ ToS **availability** in the four selected languages – (ToS in German identified for 88 out of 142 ToS, Italian and Polish versions respectively found for 79 and 55 out of these 142 ToS)
 - ❑ Possibility of identifying a **correspondence** between the different versions, given their publication date
 - ❑ Structure **similarities** (e.g. number of clauses, sections, etc.)
- Focus on 9 (potentially) unfair categories

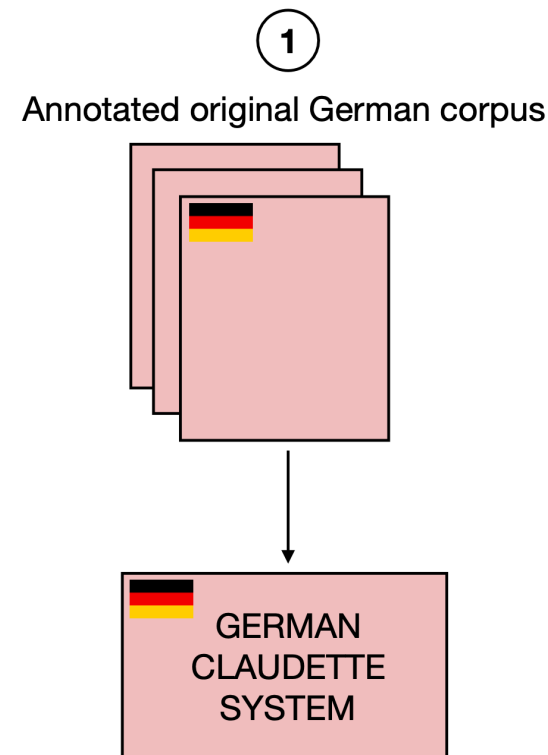
	EN	IT	DE	PL
a1	3	4	3	4
a2	48	47	40	56
a3	16	16	15	19
ch2	224	235	220	224
cr2	60	57	58	60
cr3	45	53	43	46
j1	44	48	46	46
j3	95	87	79	84
law1	53	56	55	54
law2	82	80	72	73
ltd1	63	56	64	60
ltd2	490	478	466	488
ltd3	8	6	7	8
pinc2	36	34	31	37
ter2	193	198	185	190
ter3	92	97	88	92
use2	119	117	113	123

Table 1: Corpus statistics: we report the number of annotated clauses for each tag, across the four different languages. Suffices 1, 2, and 3 represent levels of fairness: 1 means clearly fair, 2 stands for potentially unfair, and finally 3 for clearly unfair. We remark that a clause can be annotated with multiple labels.

Multilingualism: the German, Polish and Italian Claudette

Approach 1: Novel corpus for any target language

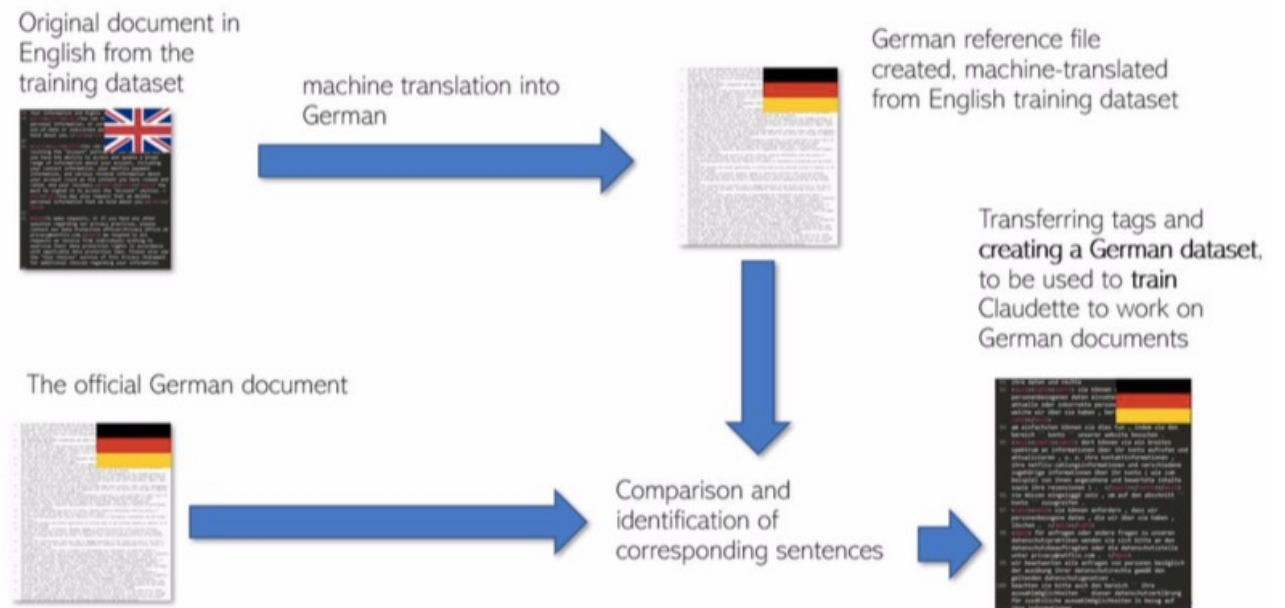
- Domain experts are required to annotate novel corpora
- Claudette is retrained from scratch
- A solution completely independent from the Claudette English Version



Multilingualism: the German, Polish and Italian Claudette

Approach 2: Semi-automated creation of a target-language dataset through annotation

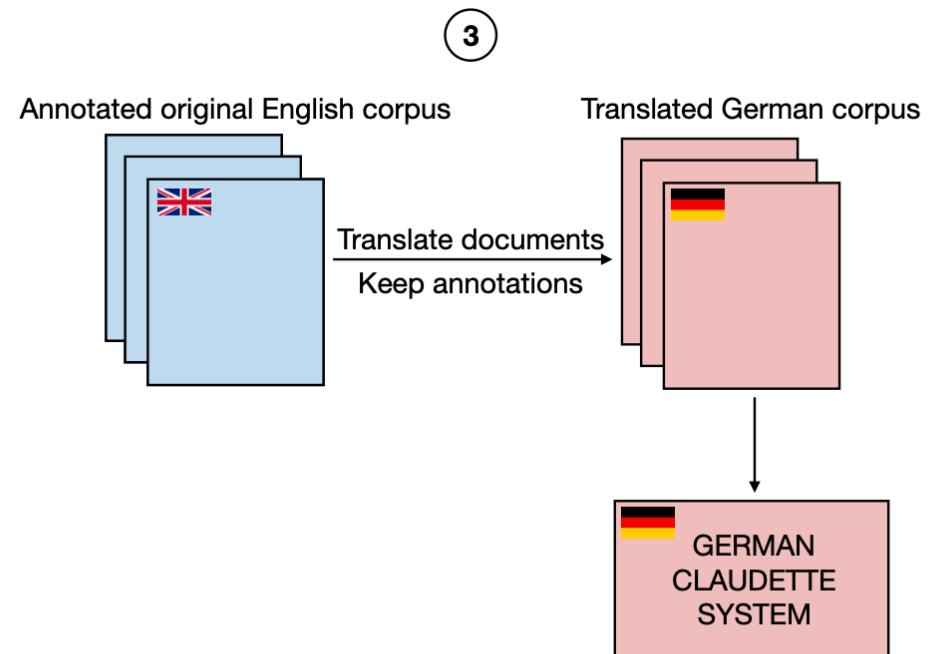
- Collect the same contracts across the languages
- Perform a projection of annotations from the English version of each document onto the corresponding document in the target language
- Re-train Claudette from scratch



Multilingualism: the German, Polish and Italian Claudette

Approach 3: Training set translation to target language.

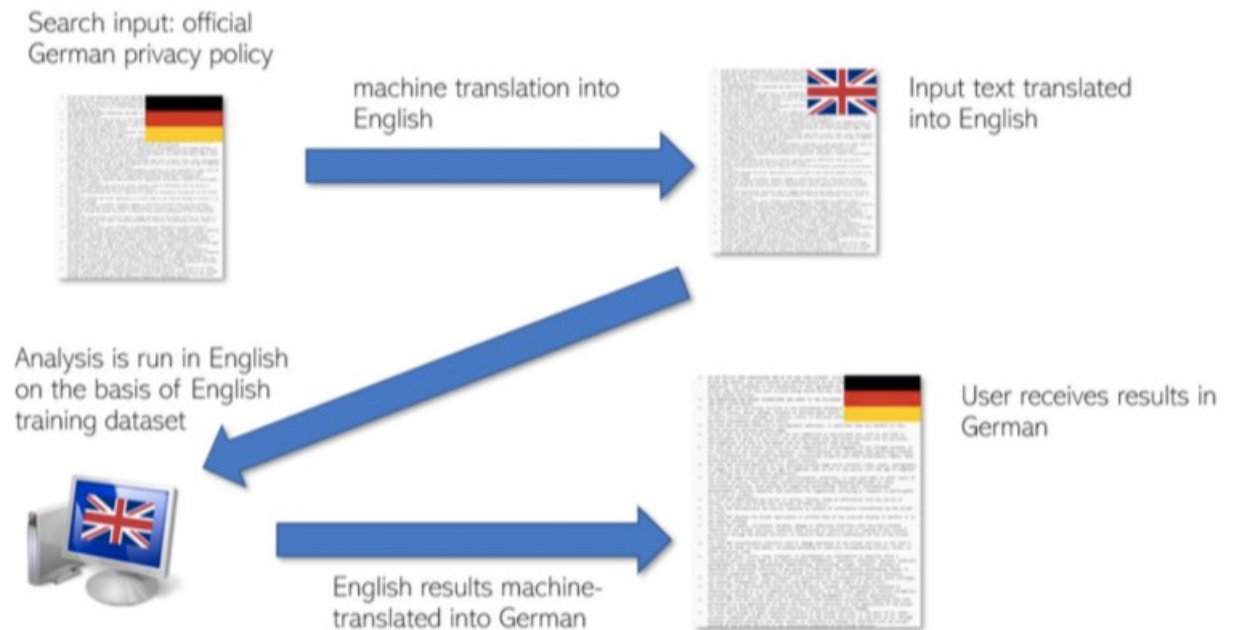
- Original Eng documents translated into the target languages
- Original annotations directly attached to translated sentences
- Claudette is retrained from scratch
- No time-consuming activity of annotating a novel corpus



Multilingualism: the German, Polish and Italian Claudette

Approach 4 – Machine translation of queries- Test set translation

- No need to re-train any novel machine learning system
- It relies on the translation of test documents into English
- The translated documents x are then classified using the Claudette English version
- Then predictions are associated back to the original documents



Multilingualism: the German, Polish and Italian Claudette

- **Results** across the three languages consistently indicate that the **fourth scenario (i.e.,** keeping only the Eng version of the machine learning system, relying on the translation of the queries at test time, is the best solution when the translation system has a high quality
- Google translate vs Apache Joshua **!!!**
- ...and the winner is... Google translate

CLAUDETTE meets GDPR

Moving to **privacy policies**: **what is different?**

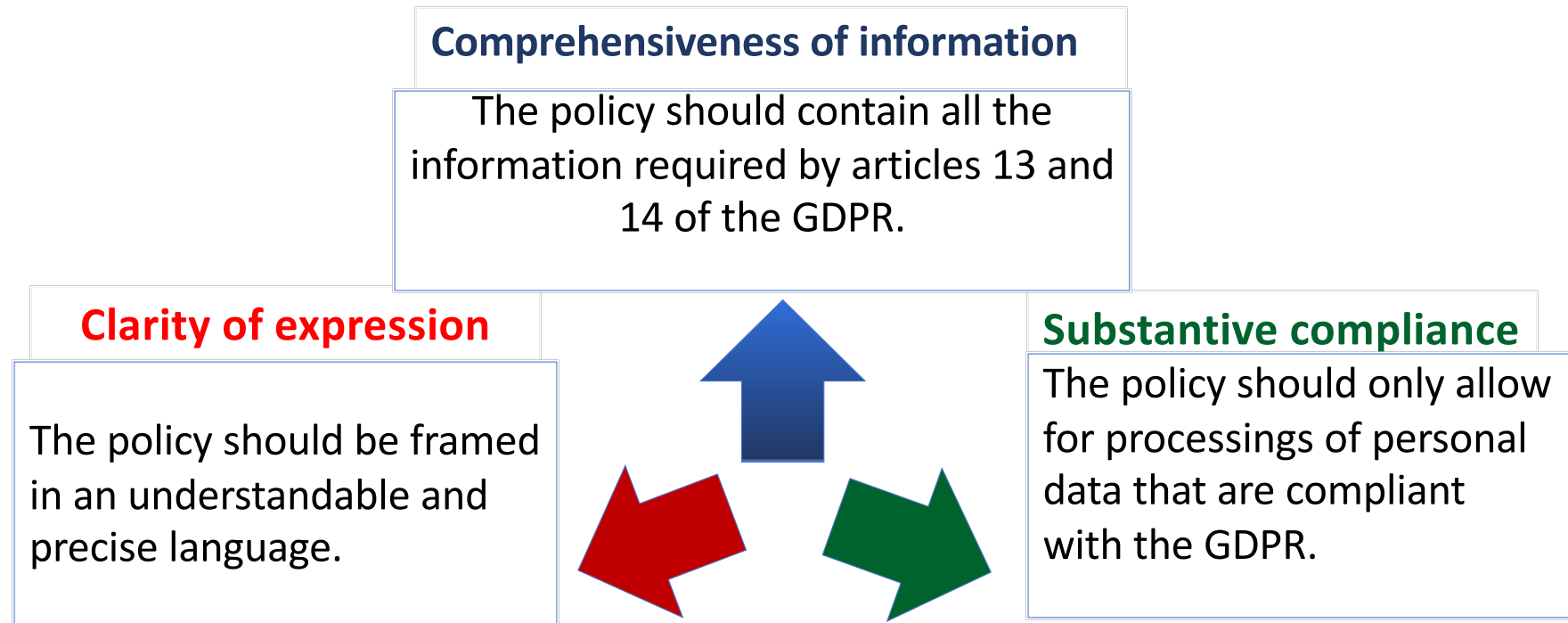
Ensure some information is **present** and **complete** (i.e., compliant with articles 13-14 of GDPR)

Detect problematic clauses for **data processing**

Detect **vague** language

CLAUDETTE meets GDPR

The Golden Standard: Lawfulness Fairness Transparency



Different Levels of Achievement: Optimal and Suboptimal (questionable or insufficient)

Examples of failure

Failure under the substantive dimension

Epic games Privacy Policy (last updated on 24 May 2018)

<cuse3> when you use our websites, games, game engines, and applications, you agree to our collection, use, disclosure, and transfer of information as described in this policy, so please review it carefully.</cuse3>

Rationale

The clause above is an unfair processing clause since it states that the data subject consents to the collection, use, disclosure and transfer of his/her information, and thus s/he is bound by the privacy policy, simply by using the Epic Games web-sites, games, game engines and applications.

Failure under the comprehensiveness dimension

Facebook Privacy Policy (last updated on 19 April 2018)

<dpo2>Contact the Data Protection Officer for Facebook Ireland Ltd.</dpo2>

Rationale

The clause above fails to be fully informative since it generically refers to the possibility of contacting the DPO but does not provide the DPO name and a postal address, only a link to an online form. Thus, it only reaches a low standard for the clarity and accessibility of the information.



Comprehensiveness of information

23 categories (GDPR art 13 and 14)

Type of required information	Symbol
Identity of the controller (controller's representative)	<id>
Contact details of the controller (controller's representative)	<contact>
Contact details of the data protection officer	<dpo>
Purposes of the processing	<purp>
Legal Basis for the processing	<basis>
Categories of personal data concerned	<cat>
Recipients or categories of recipients of the personal data	<recep>
Period for which the personal data will be stored, or the criteria used to determine that period	<ret>
Right to lodge a complaint with a supervisory authority	<complain>
...	<...>

Substantive compliance

10 categories (GDPR art 5, 6, 9 and others)

Type of clause	Symbol
Processing of special categories of personal data (e.g. health, sex life, political opinions, religious beliefs, etc.)	<sens>
Consent by using	<cuse>
Take or leave it approach	<tol>
Third party data transfers	<tp>
Policy change	<pch>
Transfer of data to third countries	<cross>
Processing of children's data	<child>
Licensing data	<lic>
Advertising	<ad>
Any other type of consent	<c>

Clarity of expression

(GDPR art. 5(1)(a), 12 (1) and others)

Is the privacy policy framed in an understandable and precise language?

4 main indicators of vagueness

Indicator	Language qualifiers
1. Conditional Terms The performance of a stated action or activity is dependent on a variable trigger	Depending, as necessary, as appropriate, as needed, otherwise reasonably, sometimes, from time to time, etc.
Example	Rationale
<vag> We also may share your information if we believe, in our sole discretion, that such disclosure is necessary :... "</vag>	The practice described as “necessary” suggests that the sharing will only occur in exceptional cases, however the clause fails to specify under what exceptional conditions the provider will disclose the information.

Clarity of expression

Indicator	Language qualifiers
2. Generalization: i.e. terms that vaguely abstract information practices using contexts that are unclear. Action(s)/Information Types are vaguely abstracted with unclear conditions.	generally, mostly, widely, general, commonly, usually, normally, typically, largely, often, primarily, among other things, etc.
Example	Rationale
<code><vag> We typically or generally collect information ..</vag></code> <code><vag> When you use an Application on a Device, we will collect and use information about you in generally similar ways and for similar purposes as when you use the TripAdvisor website.</vag></code>	The use of the generalization term “generally” obscures for the data subject the service provider activities, since it provides a large flexibility to the service provider.

Clarity of expression

(GDPR art. 5(1)(a), 12 (1) and others)

Indicator	Language qualifiers
3. Modality: it includes modal verbs, adverbs and non-specific adjectives, which create uncertainty with respect to actual action; it includes whether an action is possible. Modality does not include whether an action and/or activity is permitted. Modality mainly refers to ambiguous possibility of action or event.	may, might, could, would, possible, possibly, etc.
Example	Rationale
<vag>We may use your personal data to develop new services</vag>	it is unclear whether or not the controller will use the data subject information to develop new services and in what cases and under

Clarity of expression

Indicator	Language qualifiers
4. Non specific Numeric quantifiers: which create ambiguity as to the actual measure	certain, numerous, some, most, many, various, including (but not limited to), variety
Example	Rationale
<code><vag>When you create an Apple ID, apply for commercial credit, purchase a product, download a software update, register for a class at an Apple Retail Store, connect to our services, contact us or participate in an online survey, we may collect a variety of information, including your name, mailing address, phone number, email address, contact preferences, device identifiers, IP address, location information and credit card information.</vag></code>	it creates ambiguity with regard to the actual measure of information the data controller collect

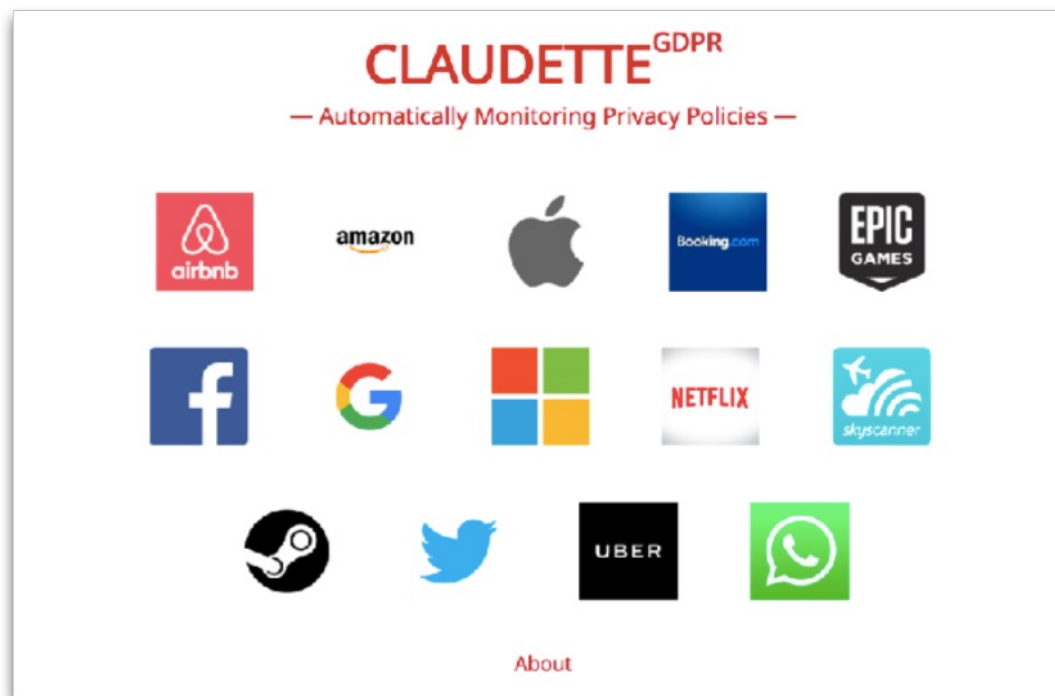
Clarity of expression

A combination of different forms of vagueness

<vag>We **generally may** share personal information we collect with **certain** service providers, **some** of whom **may** use information for their own purposes **as necessary**.</vag>

CLAUDETTE FOR GDPR

<http://claudette-gdpr.eu>



Manually annotated corpus

WHERE DID WE START?

- 14 documents (32 now)
- 3,658 sentences
- 80,398 words
- 11.0% sentences contain unclear language
- 33.9% sentences contain potentially unlawful clauses

CLAUDETTE FOR GDPR

<http://claudette-gdpr.eu>

CLAUDETTE^{GDPR}

— Automatically Monitoring Privacy Policies —

Use the checkboxes below to choose the sentences you want to highlight.
In parenthesis, you can see the number of detected sentences for each category.
By hovering your cursor over each unfair sentence, you can see the categories of each detected sentence.
If you want to see the entire policy, click the grey [...] symbol; otherwise you will see just the detected clauses.

Apple Privacy Policy

(153 sentences, 3643 words, 20861 characters)

☒ Full Information (6) ☒ Insufficient Information (19) ☒ Unclear Language (12) ☒ Problematic Processing (24)

[...]

You may be asked to provide your personal information anytime you are in contact with Apple or an Apple affiliated company .

Apple and its affiliates may share this personal information with each other and use it consistent with this Privacy Policy .

They may also combine it with other information to provide and

You are not required to provide the personal information that we

our products or services or respond to any queries you may have

[...]

When you share your content with family and friends using Apple products , send gift certificates and products , or invite others to participate in Apple services or forums , Apple may collect the information you provide about those people such as name , mailing address , email address , and phone number .

Apple will use such information to fulfill your requests , provide the relevant product or service , or for anti-fraud purposes .

In certain jurisdictions , we may ask for a government issued ID in limited circumstances including when setting up a wireless account and activating your device , for the purpose of extending commercial credit , managing reservations , or as required by law .

[...]

The personal information we collect allows us to keep you posted on Apple 's latest product announcements , software updates , and upcoming events .

If you do not want to be on our mailing list , you can opt out anytime by updating your preferences .

We also use personal information to help us create , develop , operate , deliver , and improve our products , services , content and advertising , and for loss prevention and anti-fraud purposes .

Automated tagging prototype

<http://155.185.228.137/claurette4gdpr/>

CLAUDETTE^{GDPR}

Detecting Unclear Language in Privacy Policies (beta version)

Copy your text here

Submit

[About](#)

[Cite](#)

[Contact](#)

Accuracy: 68%

Automated tagging prototype

<http://155.185.228.137/claurette4gdpr/>

CLAUDETTE^{GDPR}

Detecting Unclear Language in Privacy Policies (beta version)

Claurette found 56 clauses with unclear language (displayed in **bold**) out of 229 sentences.

[...]

Learn more: Personal information we collect We collect personal information from you and any devices (including mobile devices) you use when you: use our Services, register for an account with us, provide us information on a web form, update or add information to your account, participate in our discussion chat, or when you otherwise correspond with us.

Unclear Language unfair clause

Some of this personal information, such as a way to identify you, is necessary to enter into our User Agreement.

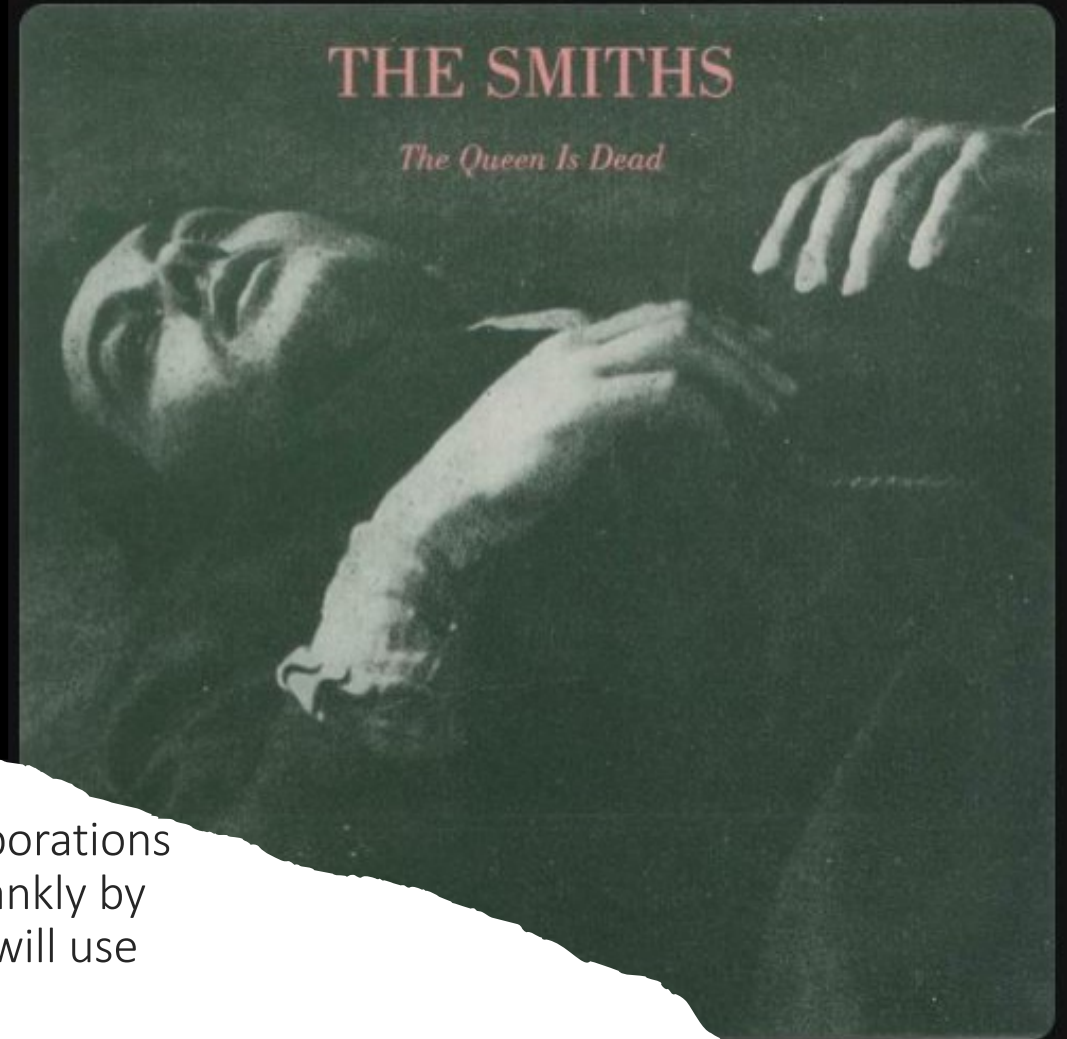
The provision of all other personal information is voluntary, but may be necessary in order to use our Services, such as the bidding, buying or selling information needed to conclude a transaction.

We may also collect personal information from other sources, as described below.

Personal information you give us when you use our Services or register for an account with us Identifying information such as your name, addresses, telephone numbers or email addresses when you register for an account with us Bidding, buying, or selling information you provide during a transaction, or other transaction-based content that you generate or that is connected to your account as a result of a transaction you are involved in Other content that you generate, or that is connected to your account (such as adding items to your basket, adding items to your Watch List, creating collections, and following other collections and sellers) Financial information (such as credit card or bank account numbers) in connection

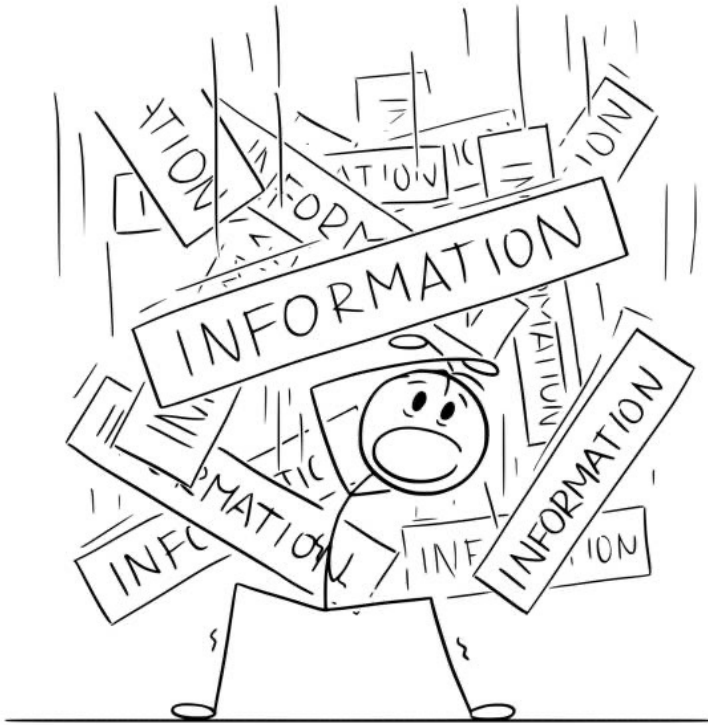


The Queen Is Dead



Hey SKYNET, could you tell me the names of all corporations who will know that I just listened to Frankly, Mr. Shankly by The Smiths and list all the purposes for which they will use this data?

The Privacy Policy Landscape



- very long and complex
- do not contain the necessary info
- are full of vague terms (“we collect data about your use of our service”) AND open-ended catalogs (“such as,” “including,” “for example”)
- do not specify what categories of data are shared with whom (“we share your personal data with our marketing partners”)

Legal Requirements for PPs



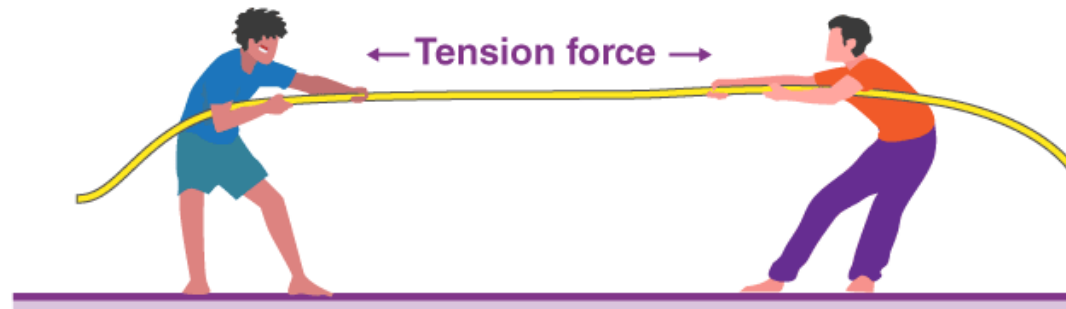
Comprehensiveness vs. Comprehensibility

- Comprehensiveness:

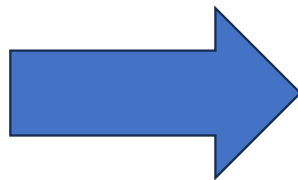
A PP should contain all the info relevant for pondering whether to use a service.

- Comprehensibility

A PP should be written in simple language and easy to understand.



WHAT IF



Assessing Privacy Policies

9 simple questions a consumer should be able to get a clear answer to:

Q1: What data does the company process about me?

Q2: For what purposes does the company use my email address?

Q3: Who does the company share my geolocation with?

Q4: What types of data are processed on the basis of consent, and for what purposes?

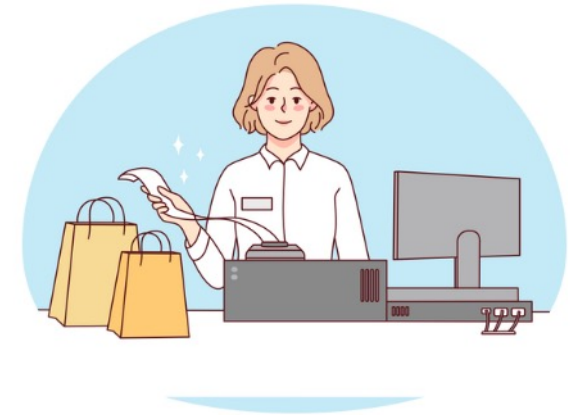
Q5: What data does the company share with Facebook?

Q6: Does the company share my data with insurers?

Q7: What categories of data does the company collect about me automatically?

Q8: How can I contact the company if I want to exercise my rights?

Q9: How long does the company keep my delivery address?



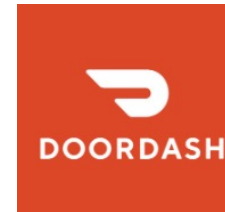
Assessing Privacy Policies: Scenarios

- Scenario 1: Human Evaluation of Existing Privacy Policies
- Scenario 2: LLMs and Mock Privacy Policy
- Scenario 3: LLMs and Real Privacy Policies



Scenario 1: Human Evaluation of Existing Privacy Policies

- **Goal:** to what extent legal experts are able to answer the 9 questions from current PPs?
- 5 PPs from food delivery sector (Deliveroo, DoorDash, Glovo, Just Eat, and Wolt)
- Length: 4'379 words (Deliveroo)- 13'765 words (Glovo)
- Each policy independently evaluated by 2 legal experts using the 9 mentioned questions
- **PP Evaluation Criteria:**
 - a) unambiguous (clear) information (i.e., definite answer could be given)
 - b) ambiguous (impossible to know) information (i.e., no definite answer could be provided)
- Each expert were required to add justifications for the evaluations



Scenario 1:Results

Question	Just Eat	DoorDash	Wolt	Glovo	Deliveroo
Q1	✗	✗	✗	✗	✗
Q2	✗	✗	✗	✗	✗
Q3	✗	✗	✗	✗	✗
Q4	✗	✗	✗	✗	✗
Q5	✗	✗	✗	✗	✗
Q6	✓	✗	✓	✓	✓
Q7	✗	✗	✗	✗	✗
Q8	✓	✓	✓	✓	✓
Q9	✗	✗	✗	✗	✗

Table 1: Legal expert evaluation of privacy policies.

The Legal Proposal

Fully Comprehensive Privacy Policies

The law should require corporations to disclose *fully comprehensive privacy policies* ideally in a *standardized form* *relax the comprehensibility requirement.*

The non ambiguity test: if it is reasonably possible for a reader to ask “**but what do you mean by category X** (e.g., usage or payment information)?”, that category should either **be split** into smaller categories **or exhaustively defined** (e.g., the amount of time you spend on the app and the article you repost; bank account and PayPal email)



The Mock PP: Orderoo INC.

Each paragraph should contain:

- A. data category
- B. source of data
- C. purpose of processing + explanation
- D. legal basis + explanation
- E. storage period
- F. sharing
 - i. the recipient's identity
 - ii. their role (processor and controller)
 - iii. the purpose of sharing
 - iv. legal basis

6880 words long

THIS IS A MOCK POLICY OF "ORDEROO INC."
A COMPANY SIMILAR TO DELIVEROO/ DOORDASH /JUST EAT/ UBER EATS etc.

ORDEROO INC. PRIVACY POLICY

This is a privacy policy of Orderoo Inc., a company located at 1 Name Street, 40121, Bologna, Italy. In this document, we explain what personal data we collect when you are using our service, what source we collect it from, for what purposes we use it, with whom we share it, and based on what legal basis.

We have appointed a Data Protection Officer, who can be contacted at dpo@orderoo.com. You can also contact us by writing to privacy@orderoo.com or at the physical address of our location.

When you use our services, we process the following categories of personal data:

1. Your email address. You provide us with your email address when registering for the service. We use your email address for the following purposes: unique identifier, it serves as a unique identifier allowing you to set up and log in to your account (contractual necessity); account access, to let you reset your password if you forget it (contractual necessity); transaction-related-communication, to send you receipts of your orders (legal obligation: to issue receipts, according to the Receipts Act); distribution of own advertising, to send you advertisements of our own services, new functionalities or new order options (legitimate interest: informing the consumers about the available offers and features, and promoting them); distribution of third-party marketing, to send you advertisements of vendors selling their products on our site (legitimate interest: to subsidize the price of the service with payments from the vendors we promote); tracking transaction history, we keep it as a part of your order history in case it becomes necessary to reveal it to investigative authorities (legal obligation: Accounting Act and Code of

A sample from the fully
comprehensive mock PP.

Scenario 2: LLMs and Mock Privacy Policy

- **Goal:** to what extent LLMs are able to answer the 9 questions from a fully comprehensive privacy policy?.
- **Tested LLMs:** GPT-4, Llama-7B, Llama-13B, Llama-70B, Mistral-7B, and Momo-70B (Issue: context window and number of tokens)
- **Selected LLMs:** GPT-4 and Llama2-7b
- **5 iterations** to assess variability across different runs
- Legal expert evaluation of the LLM answers against the Mock PP

The experimental set up

Prompt: In answering the questions please rely solely on the information included in the text and not your knowledge from other sources; please read the document carefully and mention everything, do not omit any information included in the text; please do not shorten or simplify the answers.

Scenario 2: GPT-4 Results on Mock PP



	Precision	Recall	F1	
Q1	100.0	100.0	100.0	
Q2	100.0	100.0	100.0	
Q3	100.0	100.0	100.0	
Q4	96.7	92.0	94.3	Data processed based on consent
Q5	94.4	90.0	92.1	Data shared with facebook
Q6	100.0	100.0	100.0	
Q7	100.0	78.2	87.8	Data collected automatically
Q8	100.0	100.0	100.0	
Q9	100.0	100.0	100.0	

- GPT answered the majority of questions correctly (33 of 45 questions)
- It followed the prompt instructions carefully, i.e., it used only the information available in the text and did not shorten or simplify the answers

Scenario 2: Llama2-7B Results on Mock PP

Llama2-7B provided only one (!) correct answer for the 45 questions
(9 questions for 5 runs) => impossible to use standard metrics

Error analysis:

- FN (Missing Info) = 91%
- HAL (Answers not based on the analysed policy) = 62%
- SUM = 27%
- FP (information present in the policy but irrelevant for question) = 38%
- INF (inferences) = 13%

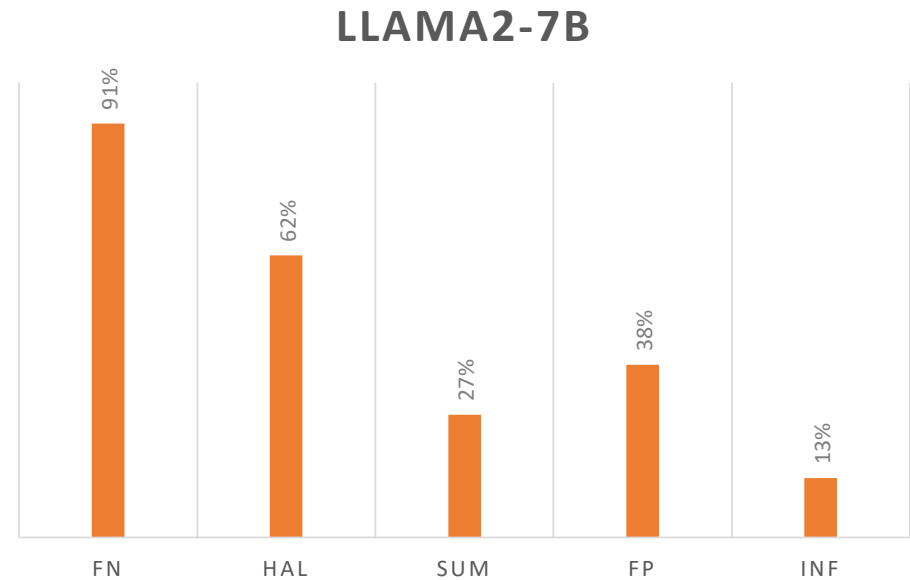
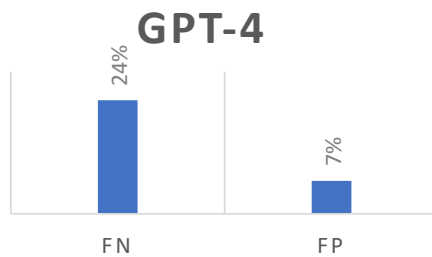


Scenario 2: GPT-4 vs Llama-7B

A comparison

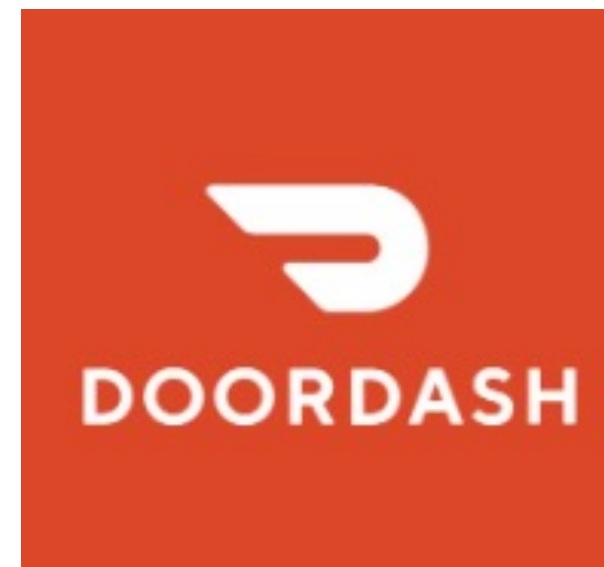
Correct answers: GPT-4=33 vs Llama2-7B=1

Error comparison



Scenario 3: LLMs and Real Privacy Policies

- **Goal:** to what extent LLMs are able to answer the 9 questions from current privacy policies?
- 5 documents: Deliveroo, DoorDash, Glovo, Just Eat, and Wolt
- Selected LLMs: GPT-4 and Llama2-7B
- Same prompt as for the experiment on the mock policy
- Legal expert evaluation of the LLMs answers against the real PPs



Scenario 3: Results

- GPT-4 provided 9 correct answers, while Llama2-7B only 1 correct answer out of 45 questions in total
- Error analysis:
 - **FN:** GPT-4 47% vs LLama2-7B 64%
 - **FP:** GPT-4 20% vs Llama2-7B 58%
 - **HAL:** GPT-4 0% vs Llama2-7B 38%
 - **SUM:** GPT-4 24% vs LLama2-7B 11%,
 - **INF:** GPT-4 33% vs Llama2-7B 31%

Conclusions

Existing PPs fail to provide meaningful information due to the trade-off between comprehensiveness and comprehensibility

➤ Our experimental setting demonstrated:

- **Scenario 1:** legal experts are unable to answer 7/9 questions due to vague or missing info in the selected PPs
- **Scenario 2:** when given well-structured, fully informative PPs, LLMs (GPT-4 and other LLMs in the future) can provide consumers with meaningful and precise answers.
- **Scenario 3:** when faced with vague and incomplete info, GPT-4 and Llama2-7B produced a high number of FN and FP, SUM, INF, HALL.

In a world of AI-readers it is possible to fully achieve both comprehensiveness and comprehensibility.

The law can and should change!!!



WEB-CRAWLER

Developed as a tool for automatic privacy policy monitoring

Two types of monitoring:

- Checking the date on the document
- Comparison of the content with the previously saved version

Earnings reports by e-mail

What we are working on

- New categories of clauses for privacy policies
- Release of the Claudette Multilingualism platform (German, Polish and Italian versions of Claudette + 4 new languages)
- Empowerment through transparency
 - Linguistic transparency
 - AI Explainability (through rationales and court rulings)
- Recommendations for service providers and users